# CANONICAL ALGEBRAIZATION OF COHERENT FORMAL SYSTEMS UNDER COHOMOLOGICAL-DIMENSION-ONE DEPTH BOUNDS

THE STACKS PROJECT AUTHORS

*Editorial note.* The exactcd(A,I)=1 assumption and the depth-plus-distance condition on a pro-isomorphic coherent system remain in the statement. The principal-ideal algebraization variant is not counted separately.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. Dualizing complexes, coherent formal modules, local-cohomology finiteness from depth, adic completion, finite modules over complete rings and Mittag–Leffler limits. The source-local topology and canonical-extension criterion are retained.

*Source reference.* Original lemma 0EHA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18125–18125. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0DXD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17428–17428. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 01YB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2550–2550. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original section 0EHN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18138–18138. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 087W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10441–10441. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0EI8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18158–18158. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

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*Date*: stacks-project:algebraization.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

*Source reference.* Original lemma 031D: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3980–3980. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0BK0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14543–14543. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0BLT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14606–14606. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0BLC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14590–14590. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original theorem 0BJV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14538–14538. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0DXU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17444–17444. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original section 0DXV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17445–17445. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0BJY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14541–14541. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0GYK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21145–21145. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original section 0EKN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18243–18243. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two

concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0AW0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13703–13703. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original equation 0880: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10445–10445. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0EJ7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18192–18192. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0G41: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20111–20111. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0BD9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14307–14307. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 00MB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 817–817. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 031C: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3979–3979. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0EHQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18140–18140. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0EIQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18175–18175. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 02IZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3318–3318. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 02JU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3349–3349. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0EHM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18137–18137. Classical finite-module completion or quasi-affine sheaf fact in the completed context proof; not the canonical-extension construction.

*Source reference.* Original lemma 09B8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11786–11786. Classical finite-module completion or quasi-affine sheaf fact in the completed context proof; not the canonical-extension construction.

*Source reference.* Original lemma 05GG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7060–7060. Classical finite-module completion or quasi-affine sheaf fact in the completed context proof; not the canonical-extension construction.

**Lemma 0.1.** *In Situation 0.2 let $(\mathcal{F}_n)$ be an object of $Coh(U, I\mathcal{O}_U)$. Assume*

(1) *$A$ has a dualizing complex and $cd(A, I) = 1$,*
(2) *$(\mathcal{F}_n)$ is pro-isomorphic to an inverse system $(\mathcal{F}''_n)$ of coherent $\mathcal{O}_U$-modules such that $depth(\mathcal{F}''_{n,y}) + \delta_Z^Y(y) \geq 3$ for all $y \in U \cap Y$.*

*Then $(\mathcal{F}_n)$ extends canonically to $X$, see Definition 0.8.*

**Situation 0.2.** Here $A$ is a Noetherian ring and $I \subset \mathfrak{a} \subset A$ are ideals. We set $X = \mathrm{Spec}(A)$, $Y = V(I) = \mathrm{Spec}(A/I)$, and $Z = V(\mathfrak{a}) = \mathrm{Spec}(A/\mathfrak{a})$. Furthermore $U = X \setminus Z$.

In this section we try to find conditions that guarantee an object of $Coh(U, I\mathcal{O}_U)$ is in the image of the completion functor $Coh(\mathcal{O}_U) \to Coh(U, I\mathcal{O}_U)$. See Cohomology of Schemes, Section 0EHN and Section 0EKN.

**Lemma 0.3.** *In Situation 0.2. Consider an inverse system $(M_n)$ of $A$-modules such that*

(1) *$M_n$ is a finite $A$-module,*
(2) *$M_n$ is annihilated by $I^n$,*
(3) *the kernel and cokernel of $M_{n+1}/I^n M_{n+1} \to M_n$ are $\mathfrak{a}$-power torsion.*

*Then $(\widetilde{M_n}|_U)$ is in $Coh(U, I\mathcal{O}_U)$. Conversely, every object of $Coh(U, I\mathcal{O}_U)$ arises in this manner.*

*Proof.* We omit the verification that $(\widetilde{M_n}|_U)$ is in $Coh(U, I\mathcal{O}_U)$. Let $(\mathcal{F}_n)$ be an object of $Coh(U, I\mathcal{O}_U)$. By Local Cohomology, Lemma 0BK0 we see that $\mathcal{F}_n = \widetilde{M_n}$ for some finite $A/I^n$-module $M_n$. After dividing $M_n$ by $H^0_{\mathfrak{a}}(M_n)$ we may assume $M_n \subset H^0(U, \mathcal{F}_n)$, see Dualizing Complexes, Lemma 0AW0 and the already referenced lemma. After replacing inductively $M_{n+1}$ by the inverse image of $M_n$ under the map $M_{n+1} \to H^0(U, \mathcal{F}_{n+1}) \to H^0(U, \mathcal{F}_n)$, we may assume $M_{n+1}$ maps into $M_n$. This gives an inverse system $(M_n)$ satisfying (1) and (2) such that $\mathcal{F}_n =$

$\widetilde{M_n}$. To see that (3) holds, use that $M_{n+1}/I^nM_{n+1} \to M_n$ is a map of finite $A$-modules which induces an isomorphism after applying $\widetilde{\ }$ and restriction to $U$ (here we use the first referenced lemma one more time). □

In Situation 0.2 we can study the completion functor Cohomology of Schemes, Equation (0880)

$$Coh(\mathcal{O}_U) \longrightarrow Coh(U, I\mathcal{O}_U), \quad \mathcal{F} \longmapsto \mathcal{F}^\wedge \tag{0.3.1}$$

If $A$ is $I$-adically complete, then this functor is fully faithful on suitable subcategories by our earlier work on algebraization of formal sections, see Section 0EKN and Lemma 0EJ7 for some sample results. Next, let $(\mathcal{F}_n)$ be an object of $Coh(U, I\mathcal{O}_U)$. Still assuming $A$ is $I$-adically complete, we can ask: When is $(\mathcal{F}_n)$ in the essential image of the completion functor displayed above?

**Lemma 0.4.** *In Situation 0.2 let $(\mathcal{F}_n)$ be an object of $Coh(U, I\mathcal{O}_U)$. Consider the following conditions:*

1. *$(\mathcal{F}_n)$ is in the essential image of the functor (0.3.1),*
2. *$(\mathcal{F}_n)$ is the completion of a coherent $\mathcal{O}_U$-module,*
3. *$(\mathcal{F}_n)$ is the completion of a coherent $\mathcal{O}_V$-module for $U \cap Y \subset V \subset U$ open,*
4. *$(\mathcal{F}_n)$ is the completion of the restriction to $U$ of a coherent $\mathcal{O}_X$-module,*
5. *$(\mathcal{F}_n)$ is the restriction to $U$ of the completion of a coherent $\mathcal{O}_X$-module,*
6. *there exists an object $(\mathcal{G}_n)$ of $Coh(X, I\mathcal{O}_X)$ whose restriction to $U$ is $(\mathcal{F}_n)$.*

*Then conditions (1), (2), (3), (4), and (5) are equivalent and imply (6). If $A$ is $I$-adically complete then condition (6) implies the others.*

*Proof.* Parts (1) and (2) are equivalent, because the completion of a coherent $\mathcal{O}_U$-module $\mathcal{F}$ is by definition the image of $\mathcal{F}$ under the functor (0.3.1). If $V \subset U$ is an open subscheme containing $U \cap Y$, then we have

$$Coh(V, I\mathcal{O}_V) = Coh(U, I\mathcal{O}_U)$$

since the category of coherent $\mathcal{O}_V$-modules supported on $V \cap Y$ is the same as the category of coherent $\mathcal{O}_U$-modules supported on $U \cap Y$. Thus the completion of a coherent $\mathcal{O}_V$-module is an object of $Coh(U, I\mathcal{O}_U)$. Having said this the equivalence of (2), (3), (4), and (5) holds because the functors $Coh(\mathcal{O}_X) \to Coh(\mathcal{O}_U) \to Coh(\mathcal{O}_V)$ are essentially surjective. See Properties, Lemma 0G41.

It is always the case that (5) implies (6). Assume $A$ is $I$-adically complete. Then any object of $Coh(X, I\mathcal{O}_X)$ corresponds to a finite $A$-module by Cohomology of Schemes, Lemma 087W. Thus we see that (6) implies (5) in this case. □

**Example 0.5.** Let $k$ be a field. Let $A = k[x, y][[t]]$ with $I = (t)$ and $\mathfrak{a} = (x, y, t)$. Let us use notation as in Situation 0.2. Observe that $U \cap Y = (D(x) \cap Y) \cup (D(y) \cap Y)$ is an affine open covering. For $n \geq 1$ consider the invertible module $\mathcal{L}_n$ of $\mathcal{O}_U/t^n\mathcal{O}_U$ given by glueing $A_x/t^nA_x$ and $A_y/t^nA_y$ via the invertible element of $A_{xy}/t^nA_{xy}$ which is the image of any power series of the form

$$u = 1 + \frac{t}{xy} + \sum_{n \geq 2} a_n \frac{t^n}{(xy)^{\varphi(n)}}$$

with $a_n \in k[x, y]$ and $\varphi(n) \in \mathbf{N}$. Then $(\mathcal{L}_n)$ is an invertible object of $Coh(U, I\mathcal{O}_U)$ which is not the completion of a coherent $\mathcal{O}_U$-module $\mathcal{L}$. We only sketch the argument and we omit most of the details. Let $y \in U \cap Y$. Then the completion of the stalk $\mathcal{L}_y$ would be an invertible module hence $\mathcal{L}_y$ is invertible. Thus there would exist an open $V \subset U$ containing $U \cap Y$ such that $\mathcal{L}|_V$ is invertible. By Divisors, Lemma 0BD9 we find an invertible $A$-module $M$ with $\widetilde{M}|_V \cong \mathcal{L}|_V$. However the

ring $A$ is a UFD hence we see $M \cong A$ which would imply $\mathcal{L}_n \cong \mathcal{O}_U/I^n\mathcal{O}_U$. Since $\mathcal{L}_2 \not\cong \mathcal{O}_U/I^2\mathcal{O}_U$ by construction we get a contradiction as desired.

Note that if we take $a_n = 0$ for $n \geq 2$, then we see that $\lim H^0(U, \mathcal{L}_n)$ is nonzero: in this case we the function $x$ on $D(x)$ and the function $x + t/y$ on $D(y)$ glue. On the other hand, if we take $a_n = 1$ and $\varphi(n) = 2^n$ or even $\varphi(n) = n^2$ then the reader can show that $\lim H^0(U, \mathcal{L}_n)$ is zero; this gives another proof that $(\mathcal{L}_n)$ is not algebraizable in this case.

If in Situation 0.2 the ring $A$ is not $I$-adically complete, then Lemma 0.4 suggests the correct thing is to ask whether $(\mathcal{F}_n)$ is in the essential image of the restriction functor

$$Coh(X, I\mathcal{O}_X) \longrightarrow Coh(U, I\mathcal{O}_U)$$

However, we can no longer say that this means $(\mathcal{F}_n)$ is algebraizable. Thus we introduce the following terminology.

**Definition 0.6.** In Situation 0.2 let $(\mathcal{F}_n)$ be an object of $Coh(U, I\mathcal{O}_U)$. We say $(\mathcal{F}_n)$ *extends to* $X$ if there exists an object $(\mathcal{G}_n)$ of $Coh(X, I\mathcal{O}_X)$ whose restriction to $U$ is isomorphic to $(\mathcal{F}_n)$.

This notion is equivalent to being algebraizable over the completion.

**Lemma 0.7.** *In Situation 0.2 let $(\mathcal{F}_n)$ be an object of $Coh(U, I\mathcal{O}_U)$. Let $A', I', \mathfrak{a}'$ be the $I$-adic completions of $A, I, \mathfrak{a}$. Set $X' = \mathrm{Spec}(A')$ and $U' = X' \setminus V(\mathfrak{a}')$. The following are equivalent*

1. *$(\mathcal{F}_n)$ extends to $X$, and*
2. *the pullback of $(\mathcal{F}_n)$ to $U'$ is the completion of a coherent $\mathcal{O}_{U'}$-module.*

*Proof.* Recall that $A \to A'$ is a flat ring map which induces an isomorphism $A/I \to A'/I'$. See Algebra, Lemmas 00MB and 031C. Thus $X' \to X$ is a flat morphism inducing an isomorphism $Y' \to Y$. Thus $U' \to U$ is a flat morphism which induces an isomorphism $U' \cap Y' \to U \cap Y$. This implies that in the commutative diagram

$$\begin{array}{ccc} Coh(X', I\mathcal{O}_{X'}) & \longrightarrow & Coh(U', I\mathcal{O}_{U'}) \\ \uparrow & & \uparrow \\ Coh(X, I\mathcal{O}_X) & \longrightarrow & Coh(U, I\mathcal{O}_U) \end{array}$$

the vertical functors are equivalences. See Cohomology of Schemes, Lemma 0EHQ. The lemma follows formally from this and the results of Lemma 0.4. □

In Situation 0.2 let $(\mathcal{F}_n)$ be an object of $Coh(U, I\mathcal{O}_U)$. To figure out if $(\mathcal{F}_n)$ extends to $X$ it makes sense to look at the $A$-module

$$M = \lim H^0(U, \mathcal{F}_n) \tag{0.7.1}$$

Observe that $M$ has a limit topology which is (a priori) coarser than the $I$-adic topology since $M \to H^0(U, \mathcal{F}_n)$ annihilates $I^nM$. There are canonical maps

$$\widetilde{M}|_U \to \widetilde{M/I^nM}|_U \to \widetilde{H^0(U, \mathcal{F}_n)}|_U \to \mathcal{F}_n$$

One could hope that $\widetilde{M}$ restricts to a coherent module on $U$ and that $(\mathcal{F}_n)$ is the completion of this module. This is naive because this has almost no chance of being true if $A$ is not complete. But even if $A$ is $I$-adically complete this notion is very difficult to work with. A less naive approach is to consider the following requirement.

**Definition 0.8.** In Situation 0.2 let $(\mathcal{F}_n)$ be an object of $Coh(U, I\mathcal{O}_U)$. We say $(\mathcal{F}_n)$ *canonically extends to* $X$ if the the inverse system

$$\{\widetilde{H^0(U, \mathcal{F}_n)}\}_{n\geq 1}$$

in $QCoh(\mathcal{O}_X)$ is pro-isomorphic to an object $(\mathcal{G}_n)$ of $Coh(X, I\mathcal{O}_X)$.

We will see in Lemma 0.9 that the condition in Definition 0.8 is stronger than the condition of Definition 0.6.

**Lemma 0.9.** *In Situation 0.2 let $(\mathcal{F}_n)$ be an object of $Coh(U, I\mathcal{O}_U)$. If $(\mathcal{F}_n)$ canonically extends to $X$, then*

1. *$(\widetilde{H^0(U, \mathcal{F}_n)})$ is pro-isomorphic to an object $(\mathcal{G}_n)$ of $Coh(X, I\mathcal{O}_X)$ unique up to unique isomorphism,*
2. *the restriction of $(\mathcal{G}_n)$ to $U$ is isomorphic to $(\mathcal{F}_n)$, i.e., $(\mathcal{F}_n)$ extends to $X$,*
3. *the inverse system $\{H^0(U, \mathcal{F}_n)\}$ satisfies the Mittag-Leffler condition, and*
4. *the module $M$ in (0.7.1) is finite over the $I$-adic completion of $A$ and the limit topology on $M$ is the $I$-adic topology.*

*Proof.* The existence of $(\mathcal{G}_n)$ in (1) follows from Definition 0.8. The uniqueness of $(\mathcal{G}_n)$ in (1) follows from Lemma 0EIQ. Write $\mathcal{G}_n = \widetilde{M_n}$. Then $\{M_n\}$ is an inverse system of finite $A$-modules with $M_n = M_{n+1}/I^n M_{n+1}$. By Definition 0.8 the inverse system $\{H^0(U, \mathcal{F}_n)\}$ is pro-isomorphic to $\{M_n\}$. Hence we see that the inverse system $\{H^0(U, \mathcal{F}_n)\}$ satisfies the Mittag-Leffler condition and that $M = \lim M_n$ (as topological modules). Thus the properties of $M$ in (4) follow from Algebra, Lemmas 09B8, 031D, and 05GG. Since $U$ is quasi-affine the canonical maps

$$\widetilde{H^0(U, \mathcal{F}_n)}|_U \to \mathcal{F}_n$$

are isomorphisms (Properties, Lemma 0EHM). We conclude that $(\mathcal{G}_n|_U)$ and $(\mathcal{F}_n)$ are pro-isomorphic and hence isomorphic by Lemma 0EIQ. □

## 1. A distance function

Let $Y$ be a Noetherian scheme and let $Z \subset Y$ be a closed subset. We define a function

$$\delta_Z^Y = \delta_Z : Y \longrightarrow \mathbf{Z}_{\geq 0} \cup \{\infty\} \tag{1.0.1}$$

which measures the "distance" of a point of $Y$ from $Z$. For an informal discussion, please see Remark 1.3. Let $y \in Y$. We set $\delta_Z(y) = \infty$ if $y$ is contained in a connected component of $Y$ which does not meet $Z$. If $y$ is contained in a connected component of $Y$ which meets $Z$, then we can find $k \geq 0$ and a system

$$V_0 \subset W_0 \supset V_1 \subset W_1 \supset \ldots \supset V_k \subset W_k$$

of integral closed subschemes of $Y$ such that $V_0 \subset Z$ and $y \in W_k$ is the generic point. Set $c_i = \text{codim}(V_i, W_i)$ for $i = 0, \ldots, k$ and $b_i = \text{codim}(V_{i+1}, W_i)$ for $i = 0, \ldots, k-1$. For such a system we set

$$\delta(V_0, W_0, V_1, \ldots, W_k) = k + \max_{i=0,1,\ldots,k} (c_i + c_{i+1} + \ldots + c_k - b_i - b_{i+1} - \ldots - b_{k-1})$$

This is $\geq k$ as we can take $i = k$ and we have $c_k \geq 0$. Finally, we set

$$\delta_Z(y) = \min \delta(V_0, W_0, V_1, \ldots, W_k)$$

where the minimum is over all systems of integral closed subschemes of $Y$ as above.

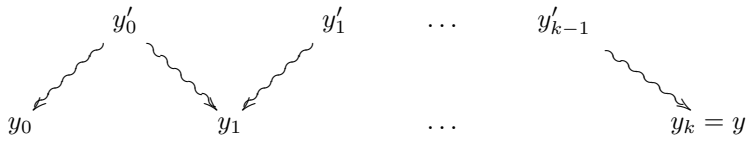
**Lemma 1.1.** *Let $Y$ be a Noetherian scheme and let $Z \subset Y$ be a closed subset.*

1. *For $y \in Y$ we have $\delta_Z(y) = 0 \Leftrightarrow y \in Z$.*
2. *The subsets $\{y \in Y \mid \delta_Z(y) \leq k\}$ are stable under specialization.*

(3) *For $y \in Y$ and $z \in \overline{\{y\}} \cap Z$ we have $\dim(\mathcal{O}_{\overline{\{y\}},z}) \geq \delta_Z(y)$.*
(4) *If $\delta$ is a dimension function on $Y$, then $\delta_Z(y) \geq \delta(y) - \delta_{max}$ where $\delta_{max}$ is the maximum value of $\delta$ on $Z$.*
(5) *If $Y = \mathrm{Spec}(A)$ is the spectrum of a catenary Noetherian local ring with maximal ideal $\mathfrak{m}$ and $Z = \{\mathfrak{m}\}$, then $\delta_Z(y) = \dim(\overline{\{y\}})$.*
(6) *If $Y' \subset Y$ is an open subscheme, then $\delta^{Y'}_{Y' \cap Z}(y') \geq \delta^Y_Z(y')$ for $y' \in Y'$.*

*Assume $Y$ is catenary. Then*

(7) *Let $y' \rightsquigarrow y$ be an immediate specialization of points of $Y$. If $Y$ is catenary, then $\delta_Z(y') \leq \delta_Z(y) + 1$.*
(8) *Given a pattern of specializations*




*between points of $Y$ with $y_0 \in Z$ and $y'_i \rightsquigarrow y_i$ an immediate specialization, then $\delta_Z(y_k) \leq k$.*

*Proof.* Proof of (1). If $y \in Z$, then we can take $k = 0$ and $V_0 = W_0 = \overline{\{y\}}$ and we get $\delta(V_0, W_0) = 0$ so $\delta_Z(y) = 0$. If $y \notin Z$, then for every system $V_0 \subset W_0 \supset V_1 \subset W_1 \supset \ldots \subset W_k$ for $y$ we either have $k = 0$ and $V_0 \neq W_0$ or $k > 0$. In both cases $\delta(V_0, W_0, \ldots, W_k) > 0$. Hence $\delta_Z(y) > 0$.

Proof of (2). Let $y \rightsquigarrow y'$ be a nontrivial specialization and let $V_0 \subset W_0 \supset V_1 \subset W_1 \supset \ldots \subset W_k$ is a system for $y$. Here there are two cases. Case I: $V_k = W_k$, i.e., $c_k = 0$. In this case we can set $V'_k = W'_k = \overline{\{y'\}}$. An easy computation shows that $\delta(V_0, W_0, \ldots, V'_k, W'_k) \leq \delta(V_0, W_0, \ldots, V_k, W_k)$ because only $b_{k-1}$ is changed into a bigger integer. Case II: $V_k \neq W_k$, i.e., $c_k > 0$. In this case, setting $V_{k+1} = W_{k+1} = \overline{\{y'\}}$ we see that $V_0 \subset W_0 \supset \ldots \subset W_k \supset V_{k+1} \subset W_{k+1}$ is a system for $y'$. Then $c_{k+1} = 0$ and $b_k > 0$ so we get

$$\begin{aligned}
&\delta(V_0, \ldots, W_{k+1}) \\
&= k + 1 + \max_{i=0,1,\ldots,k+1}(c_i + c_{i+1} + \ldots + c_k + c_{k+1} - b_i - b_{i+1} - \ldots - b_{k-1} - b_k) \\
&= k + 1 + \max_{i=0,1,\ldots,k+1}(c_i + c_{i+1} + \ldots + c_k - b_i - b_{i+1} - \ldots - b_{k-1} - b_k) \\
&\leq k + \max_{i=0,1,\ldots,k}(c_i + c_{i+1} + \ldots + c_k - b_i - b_{i+1} - \ldots - b_{k-1}) \\
&= \delta(V_0, \ldots, W_k)
\end{aligned}$$

The inequality holds because $c_k > 0$ and $b_k > 0$ which in particular implies that $\delta(V_0, \ldots, W_k) \geq k + c_k \geq k + 1$.

Proof of (3). Given $y \in Y$ and $z \in \overline{\{y\}} \cap Z$ we get the system

$$V_0 = \overline{\{z\}} \subset W_0 = \overline{\{y\}}$$

and $c_0 = \mathrm{codim}(V_0, W_0) = \dim(\mathcal{O}_{\overline{\{y\}},z})$ by Properties, Lemma 02IZ. Thus we see that $\delta(V_0, W_0) = 0 + c_0 = c_0$ which proves what we want.

Proof of (4). Let $\delta$ be a dimension function on $Y$. Let $V_0 \subset W_0 \supset V_1 \subset W_1 \supset \ldots \subset W_k$ be a system for $y$. Let $y'_i \in W_i$ and $y_i \in V_i$ be the generic points, so $y_0 \in Z$ and $y_k = y$. Then we see that

$$\delta(y_i) - \delta(y_{i-1}) = \delta(y'_{i-1}) - \delta(y_{i-1}) - \delta(y'_{i-1}) + \delta(y_i) = c_{i-1} - b_{i-1}$$

Finally, we have $\delta(y'_k) - \delta(y_{k-1}) = c_k$. Thus we see that

$$\delta(y) - \delta(y_0) = c_0 + \ldots + c_k - b_0 - \ldots - b_{k-1}$$

We conclude $\delta(V_0, W_0, \ldots, W_k) \geq k + \delta(y) - \delta(y_0)$ which proves what we want.

Proof of (5). The function $\delta(y) = \dim(\overline{\{y\}})$ is a dimension function. Hence $\delta(y) \leq \delta_Z(y)$ by part (4). By part (3) we have $\delta_Z(y) \leq \delta(y)$ and we are done.

Proof of (6). This is clear as their are fewer systems to consider in the computation of $\delta^{Y'}_{Y' \cap Z}$.

Proof of (7). Let $V_0 \subset W_0 \supset V_1 \subset W_1 \supset \ldots \subset W_k$ be a system for $y$. Set $W'_k = \overline{\{y'\}}$. Since $Y$ is catenary, we see that $\operatorname{codim}(V_k, W'_k) = \operatorname{codim}(V_k, W_k) + 1$. It follows easily that $\delta(V_0, \ldots, V_k, W'_k) \leq \delta(V_0, \ldots, V_k, W_k) + 1$ which proves what we want.

Proof of (8): combine (7) and (2). □

**Lemma 1.2.** *Let $Y$ be a universally catenary Noetherian scheme. Let $Z \subset Y$ be a closed subscheme. Let $f : Y' \to Y$ be a finite type morphism all of whose fibres have dimension $\leq e$. Set $Z' = f^{-1}(Z)$. Then*

$$\delta_Z(y) \leq \delta_{Z'}(y') + e - trdeg_{\kappa(y)}(\kappa(y'))$$

*for $y' \in Y'$ with image $y \in Y$.*

*Proof.* If $\delta_{Z'}(y') = \infty$, then there is nothing to prove. If $\delta_{Z'}(y') < \infty$, then we choose a system of integral closed subschemes

$$V'_0 \subset W'_0 \supset V'_1 \subset W'_1 \supset \ldots \subset W'_k$$

of $Y'$ with $V'_0 \subset Z'$ and $y'$ the generic point of $W'_k$ such that $\delta_{Z'}(y') = \delta(V'_0, W'_0, \ldots, W'_k)$. Denote

$$V_0 \subset W_0 \supset V_1 \subset W_1 \supset \ldots \subset W_k$$

the scheme theoretic images of the above schemes in $Y$. Observe that $y$ is the generic point of $W_k$ and that $V_0 \subset Z$. For each $i$ we look at the diagram

$$\begin{array}{ccccc} V'_i & \longrightarrow & W'_i & \longleftarrow & V'_{i+1} \\ \downarrow & & \downarrow & & \downarrow \\ V_i & \longrightarrow & W_i & \longleftarrow & V_{i+1} \end{array}$$

Denote $n_i$ the relative dimension of $V'_i/V_i$ and $m_i$ the relative dimension of $W'_i/W_i$; more precisely these are the transcendence degrees of the corresponding extensions of the function fields. Set $c_i = \operatorname{codim}(V_i, W_i)$, $c'_i = \operatorname{codim}(V'_i, W'_i)$, $b_i = \operatorname{codim}(V_{i+1}, W_i)$, and $b'_i = \operatorname{codim}(V'_{i+1}, W'_i)$. By the dimension formula we have

$$c_i = c'_i + n_i - m_i \quad \text{and} \quad b_i = b'_i + n_{i+1} - m_i$$

See Morphisms, Lemma 02JU. Hence $c_i - b_i = c'_i - b'_i + n_i - n_{i+1}$. Thus we see that

$$\begin{aligned} & c_i + c_{i+1} + \ldots + c_k - b_i - b_{i+1} - \ldots - b_{k-1} \\ & = c'_i + c'_{i+1} + \ldots + c'_k - b'_i - b'_{i+1} - \ldots - b'_{k-1} + n_i - n_k + c_k - c'_k \\ & = c'_i + c'_{i+1} + \ldots + c'_k - b'_i - b'_{i+1} - \ldots - b'_{k-1} + n_i - m_k \end{aligned}$$

Thus we see that

$$\begin{aligned}
&\max_{i=0,\ldots,k}(c_i + c_{i+1} + \ldots + c_k - b_i - b_{i+1} - \ldots - b_{k-1}) \\
&\quad = \max_{i=0,\ldots,k}(c'_i + c'_{i+1} + \ldots + c'_k - b'_i - b'_{i+1} - \ldots - b'_{k-1} + n_i - m_k) \\
&\quad = \max_{i=0,\ldots,k}(c'_i + c'_{i+1} + \ldots + c'_k - b'_i - b'_{i+1} - \ldots - b'_{k-1} + n_i) - m_k \\
&\quad \leq \max_{i=0,\ldots,k}(c'_i + c'_{i+1} + \ldots + c'_k - b'_i - b'_{i+1} - \ldots - b'_{k-1}) + e - m_k
\end{aligned}$$

Since $m_k = \mathrm{trdeg}_{\kappa(y)}(\kappa(y'))$ we conclude that

$$\delta(V_0, W_0, \ldots, W_k) \leq \delta(V'_0, W'_0, \ldots, W'_k) + e - \mathrm{trdeg}_{\kappa(y)}(\kappa(y'))$$

as desired. □

*Remark* 1.3. Let $Y$ be a catenary Noetherian scheme and let $Z \subset Y$ be a closed subset. By Lemma 1.1 we have

$$\delta_Z(y) \leq \min\left\{ k \middle| \begin{array}{c} \text{there exist specializations in } Y \\ y_0 \leftarrow y'_0 \rightarrow y_1 \leftarrow y'_1 \rightarrow \ldots \leftarrow y'_{k-1} \rightarrow y_k = y \\ \text{with } y_0 \in Z \text{ and } y'_i \rightsquigarrow y_i \text{ immediate} \end{array} \right\}$$

We claim that if $Y$ is of finite type over a field, then equality holds. If we ever need this result we will formulate a precise result and prove it here. However, in general if we define $\delta_Z$ by the right hand side of this inequality, then we don't know if Lemma 1.2 remains true.

**Example 1.4.** Let $k$ be a field and $Y = \mathbf{A}^n_k$. Denote $\delta : Y \to \mathbf{Z}_{\geq 0}$ the usual dimension function.

(1) If $Z = \{z\}$ for some closed point $z$, then
   (a) $\delta_Z(y) = \delta(y)$ if $y \rightsquigarrow z$ and
   (b) $\delta_Z(y) = \delta(y) + 1$ if $y \not\rightsquigarrow z$.

(2) If $Z$ is a closed subvariety and $W = \overline{\{y\}}$, then
   (a) $\delta_Z(y) = 0$ if $W \subset Z$,
   (b) $\delta_Z(y) = \dim(W) - \dim(Z)$ if $Z$ is contained in $W$,
   (c) $\delta_Z(y) = 1$ if $\dim(W) \leq \dim(Z)$ and $W \not\subset Z$,
   (d) $\delta_Z(y) = \dim(W) - \dim(Z) + 1$ if $\dim(W) > \dim(Z)$ and $Z \not\subset W$.

A generalization of case (1) is if $Y$ is of finite type over a field and $Z = \{z\}$ is a closed point. Then $\delta_Z(y) = \delta(y) + t$ where $t$ is the minimum length of a chain of curves connecting $z$ to a closed point of $\overline{\{y\}}$.

## 2. Formal sections, III

In this section we prove Lemma 2.5 which (in the setting of Noetherian schemes and coherent modules) is the analogue of Cohomology, Lemma 0EHA in case the ideal $I$ is not assumed principal but has the property that $\mathrm{cd}(A, I) = 1$.

**Lemma 2.1.** *Let $I = (f_1, \ldots, f_r)$ be an ideal of a Noetherian ring $A$. If $cd(A, I) = 1$, then there exist $c \geq 1$ and maps $\varphi_j : I^c \to A$ such that $\sum f_j\varphi_j : I^c \to I$ is the inclusion map.*

*Proof.* Since $\mathrm{cd}(A, I) = 1$ the complement $U = \mathrm{Spec}(A) \setminus V(I)$ is affine (Local Cohomology, Lemma 0DXD). Say $U = \mathrm{Spec}(B)$. Then $IB = B$ and we can write $1 = \sum_{j=1,\ldots,r} f_j b_j$ for some $b_j \in B$. By Cohomology of Schemes, Lemma 01YB we can represent $b_j$ by maps $\varphi_j : I^c \to A$ for some $c \geq 0$. Then $\sum f_j\varphi_j : I^c \to I \subset A$ is the canonical embedding, after possibly replacing $c$ by a larger integer, by the same lemma. □

**Lemma 2.2.** *Let $I = (f_1, \ldots, f_r)$ be an ideal of a Noetherian ring $A$ with $cd(A, I) = 1$. Let $c \geq 1$ and $\varphi_j : I^c \to A$, $j = 1, \ldots, r$ be as in Lemma 2.1. Then there is a unique graded $A$-algebra map*

$$\Phi : \bigoplus\nolimits_{n\geq 0} I^{nc} \to A[T_1, \ldots, T_r]$$

*with $\Phi(g) = \sum \varphi_j(g)T_j$ for $g \in I^c$. Moreover, the composition of $\Phi$ with the map $A[T_1, \ldots, T_r] \to \bigoplus_{n\geq 0} I^n$, $T_j \mapsto f_j$ is the inclusion map $\bigoplus_{n\geq 0} I^{nc} \to \bigoplus_{n\geq 0} I^n$.*

*Proof.* For each $j$ and $m \geq c$ the restriction of $\varphi_j$ to $I^m$ is a map $\varphi_j : I^m \to I^{m-c}$. Given $j_1, \ldots, j_n \in \{1, \ldots, r\}$ we claim that the composition

$$\varphi_{j_1} \ldots \varphi_{j_n} : I^{nc} \to I^{(n-1)c} \to \ldots \to I^c \to A$$

is independent of the order of the indices $j_1, \ldots, j_n$. Namely, if $g = g_1 \ldots g_n$ with $g_i \in I^c$, then we see that

$$(\varphi_{j_1} \ldots \varphi_{j_n})(g) = \varphi_{j_1}(g_1) \ldots \varphi_{j_n}(g_n)$$

is independent of the ordering as multiplication in $A$ is commutative. Thus we can define $\Phi$ by sending $g \in I^{nc}$ to

$$\Phi(g) = \sum\nolimits_{e_1+\ldots+e_r=n} (\varphi_1^{e_1} \circ \ldots \circ \varphi_r^{e_r})(g) T_1^{e_1} \ldots T_r^{e_r}$$

It is straightforward to prove that this is a graded $A$-algebra homomorphism with the desired property. Uniqueness is immediate as is the final property. This proves the lemma. □

**Lemma 2.3.** *Let $I = (f_1, \ldots, f_r)$ be an ideal of a Noetherian ring $A$ with $cd(A, I) = 1$. Let $c \geq 1$ and $\varphi_j : I^c \to A$, $j = 1, \ldots, r$ be as in Lemma 2.1. Let $A \to B$ be a ring map with $B$ Noetherian and let $N$ be a finite $B$-module. Then, after possibly increasing $c$ and adjusting $\varphi_j$ accordingly, there is a unique unique graded $B$-module map*

$$\Phi_N : \bigoplus\nolimits_{n\geq 0} I^{nc} N \to N[T_1, \ldots, T_r]$$

*with $\Phi_N(gx) = \Phi(g)x$ for $g \in I^{nc}$ and $x \in N$ where $\Phi$ is as in Lemma 2.2. The composition of $\Phi_N$ with the map $N[T_1, \ldots, T_r] \to \bigoplus_{n\geq 0} I^n N$, $T_j \mapsto f_j$ is the inclusion map $\bigoplus_{n\geq 0} I^{nc} N \to \bigoplus_{n\geq 0} I^n N$.*

*Proof.* The uniqueness is clear from the formula and the uniqueness of $\Phi$ in Lemma 2.2. Consider the Noetherian $A$-algebra $B' = B \oplus N$ where $N$ is an ideal of square zero. To show the existence of $\Phi_N$ it is enough (via Lemma 2.1) to show that $\varphi_j$ extends to a map $\varphi_j' : I^c B' \to B'$ after possibly increasing $c$ to some $c'$ (and replacing $\varphi_j$ by the composition of the inclusion $I^{c'} \to I^c$ with $\varphi_j$). Recall that $\varphi_j$ corresponds to a section

$$h_j \in \Gamma(\mathrm{Spec}(A) \setminus V(I), \mathcal{O}_{\mathrm{Spec}(A)})$$

see Cohomology of Schemes, Lemma 01YB. (This is in fact how we chose our $\varphi_j$ in the proof of Lemma 2.1.) Let us use the same lemma to represent the pullback

$$h_j' \in \Gamma(\mathrm{Spec}(B') \setminus V(IB'), \mathcal{O}_{\mathrm{Spec}(B')})$$

of $h_j$ by a $B'$-linear map $\varphi_j' : I^{c'} B' \to B'$ for some $c' \geq c$. The agreement with $\varphi_j$ will hold for $c'$ sufficiently large by a further application of the lemma: namely we can test agreement on a finite list of generators of $I^{c'}$. Small detail omitted. □

**Lemma 2.4.** *Let $I = (f_1, \ldots, f_r)$ be an ideal of a Noetherian ring $A$ with $cd(A, I) = 1$. Let $c \geq 1$ and $\varphi_j : I^c \to A$, $j = 1, \ldots, r$ be as in Lemma 2.1. Let $X$ be a Noetherian scheme over* Spec$(A)$*. Let*

$$\ldots \to \mathcal{F}_3 \to \mathcal{F}_2 \to \mathcal{F}_1$$

*be an inverse system of coherent $\mathcal{O}_X$-modules such that $\mathcal{F}_n = \mathcal{F}_{n+1}/I^n\mathcal{F}_{n+1}$. Set $\mathcal{F} = \lim \mathcal{F}_n$. Then, after possibly increasing $c$ and adjusting $\varphi_j$ accordingly, there exists a unique graded $\mathcal{O}_X$-module map*

$$\Phi_{\mathcal{F}} : \bigoplus\nolimits_{n\geq 0} I^{nc}\mathcal{F} \longrightarrow \mathcal{F}[T_1, \ldots, T_r]$$

*with $\Phi_{\mathcal{F}}(gs) = \Phi(g)s$ for $g \in I^{nc}$ and $s$ a local section of $\mathcal{F}$ where $\Phi$ is as in Lemma 2.2. The composition of $\Phi_{\mathcal{F}}$ with the map $\mathcal{F}[T_1, \ldots, T_r] \to \bigoplus_{n\geq 0} I^n\mathcal{F}$, $T_j \mapsto f_j$ is the canonical inclusion $\bigoplus_{n\geq 0} I^{nc}\mathcal{F} \to \bigoplus_{n\geq 0} I^n\mathcal{F}$.*

*Proof.* The uniqueness is immediate from the $\mathcal{O}_X$-linearity and the requirement that $\Phi_{\mathcal{F}}(gs) = \Phi(g)s$ for $g \in I^{nc}$ and $s$ a local section of $\mathcal{F}$. Thus we may assume $X = \mathrm{Spec}(B)$ is affine. Observe that $(\mathcal{F}_n)$ is an object of the category $Coh(X, I\mathcal{O}_X)$ introduced in Cohomology of Schemes, Section 0EHN. Let $B' = B^\wedge$ be the $I$-adic completion of $B$. By Cohomology of Schemes, Lemma 087W the object $(\mathcal{F}_n)$ corresponds to a finite $B'$-module $N$ in the sense that $\mathcal{F}_n$ is the coherent module associated to the finite $B$-module $N/I^nN$. Applying Lemma 2.3 to $I \subset A \to B'$ and $N$ we see that, after possibly increasing $c$ and adjusting $\varphi_j$ accordingly, we get unique maps

$$\Phi_N : \bigoplus\nolimits_{n\geq 0} I^{nc}N \to N[T_1, \ldots, T_r]$$

with the corresponding properties. Note that in degree $n$ we obtain an inverse system of maps $N/I^mN \to \bigoplus_{e_1+\ldots+e_r=n} N/I^{m-nc}N \cdot T_1^{e_1} \ldots T_r^{e_r}$ for $m \geq nc$. Translating back into coherent sheaves we see that $\Phi_N$ corresponds to a system of maps

$$\Phi_m^n : I^{nc}\mathcal{F}_m \longrightarrow \bigoplus\nolimits_{e_1+\ldots+e_r=n} \mathcal{F}_{m-nc} \cdot T_1^{e_1} \ldots T_r^{e_r}$$

for varying $m \geq nc$ and $n \geq 1$. Taking the inverse limit of these maps over $m$ we obtain $\Phi_{\mathcal{F}} = \bigoplus_n \lim_m \Phi_m^n$. Note that $\lim_m I^t\mathcal{F}_m = I^t\mathcal{F}$ as can be seen by evaluating on affines for example, but in fact we don't need this because it is clear there is a map $I^t\mathcal{F} \to \lim_m I^t\mathcal{F}_m$. □

**Lemma 2.5.** *Let $I$ be an ideal of a Noetherian ring $A$. Let $X$ be a Noetherian scheme over $\mathrm{Spec}(A)$. Let*

$$\ldots \to \mathcal{F}_3 \to \mathcal{F}_2 \to \mathcal{F}_1$$

*be an inverse system of coherent $\mathcal{O}_X$-modules such that $\mathcal{F}_n = \mathcal{F}_{n+1}/I^n\mathcal{F}_{n+1}$. If $cd(A, I) = 1$, then for all $p \in \mathbf{Z}$ the limit topology on $\lim H^p(X, \mathcal{F}_n)$ is $I$-adic.*

*Proof.* First it is clear that $I^t \lim H^p(X, \mathcal{F}_n)$ maps to zero in $H^p(X, \mathcal{F}_t)$. Thus the $I$-adic topology is finer than the limit topology. For the converse we set $\mathcal{F} = \lim \mathcal{F}_n$, we pick generators $f_1, \ldots, f_r$ of $I$, we pick $c \geq 1$, and we choose $\Phi_{\mathcal{F}}$ as in Lemma 2.4. We will use the results of Lemma 0EI8 without further mention. In particular we have a short exact sequence

$$0 \to R^1 \lim H^{p-1}(X, \mathcal{F}_n) \to H^p(X, \mathcal{F}) \to \lim H^p(X, \mathcal{F}_n) \to 0$$

Thus we can lift any element $\xi$ of $\lim H^p(X, \mathcal{F}_n)$ to an element $\xi' \in H^p(X, \mathcal{F})$. Suppose $\xi$ maps to zero in $H^p(X, \mathcal{F}_{nc})$ for some $n$, in other words, suppose $\xi$ is "small" in the limit topology. We have a short exact sequence

$$0 \to I^{nc}\mathcal{F} \to \mathcal{F} \to \mathcal{F}_{nc} \to 0$$

and hence the assumption means we can lift $\xi'$ to an element $\xi'' \in H^p(X, I^{nc}\mathcal{F})$. Applying $\Phi_{\mathcal{F}}$ we get

$$\Phi_{\mathcal{F}}(\xi'') = \sum\nolimits_{e_1+\ldots+e_r=n} \xi'_{e_1,\ldots,e_r} \cdot T_1^{e_1} \ldots T_r^{e_r}$$

for some $\xi'_{e_1,\ldots,e_r} \in H^p(X,\mathcal{F})$. Letting $\xi_{e_1,\ldots,e_r} \in \lim H^p(X,\mathcal{F}_n)$ be the images and using the final assertion of Lemma 2.4 we conclude that

$$\xi = \sum f_1^{e_1} \ldots f_r^{e_r} \xi_{e_1,\ldots,e_r}$$

is in $I^n \lim H^p(X,\mathcal{F}_n)$ as desired. □

**Example 2.6.** Let $k$ be a field. Let $A = k[x,y][[s,t]]/(xs-yt)$. Let $I = (s,t)$ and $\mathfrak{a} = (x,y,s,t)$. Let $X = \mathrm{Spec}(A) - V(\mathfrak{a})$ and $\mathcal{F}_n = \mathcal{O}_X/I^n\mathcal{O}_X$. Observe that the rational function

$$g = \frac{t}{x} = \frac{s}{y}$$

is regular in an open neighbourhood $V \subset X$ of $V(I\mathcal{O}_X)$. Hence every power $g^e$ determines a section $g^e \in M = \lim H^0(X,\mathcal{F}_n)$. Observe that $g^e \to 0$ as $e \to \infty$ in the limit topology on $M$ since $g^e$ maps to zero in $\mathcal{F}_e$. On the other hand, $g^e \notin IM$ for any $e$ as the reader can see by computing $H^0(U,\mathcal{F}_n)$; computation omitted. Observe that $\mathrm{cd}(A,I) = 2$. Thus the result of Lemma 2.5 is sharp.

**Lemma 2.7.** *In Situation 0.2 let $(\mathcal{F}_n)$ be an object of $Coh(U, I\mathcal{O}_U)$. Let $M$ be as in (0.7.1). Assume*

(a) *the inverse system $H^0(U,\mathcal{F}_n)$ has Mittag-Leffler,*
(b) *the limit topology on $M$ agrees with the $I$-adic topology, and*
(c) *the image of $M \to H^0(U,\mathcal{F}_n)$ is a finite $A$-module for all $n$.*

*Then $(\mathcal{F}_n)$ extends canonically to $X$. In particular, if $A$ is $I$-adically complete, then $(\mathcal{F}_n)$ is the completion of a coherent $\mathcal{O}_U$-module.*

*Proof.* Since $H^0(U,\mathcal{F}_n)$ has the Mittag-Leffler condition and since the limit topology on $M$ is the $I$-adic topology we see that $\{M/I^nM\}$ and $\{H^0(U,\mathcal{F}_n)\}$ are pro-isomorphic inverse systems of $A$-modules. Thus if we set

$$\mathcal{G}_n = \widetilde{M/I^nM}$$

then we see that to verify the condition in Definition 0.8 it suffices to show that $M$ is a finite module over the $I$-adic completion of $A$. This follows from the fact that $M/I^nM$ is finite by condition (c) and the above and Algebra, Lemma 031D. □

The following is in some sense the most straightforward possible application Lemma 2.7 above.

**Lemma 2.8.** *In Situation 0.2 let $(\mathcal{F}_n)$ be an object of $Coh(U, I\mathcal{O}_U)$. Assume*

(1) *$I = (f)$ is a principal ideal for a nonzerodivisor $f \in \mathfrak{a}$,*
(2) *$\mathcal{F}_n$ is a finite locally free $\mathcal{O}_U/f^n\mathcal{O}_U$-module,*
(3) *$H^1_\mathfrak{a}(A/fA)$ and $H^2_\mathfrak{a}(A/fA)$ are finite $A$-modules.*

*Then $(\mathcal{F}_n)$ extends canonically to $X$. In particular, if $A$ is complete, then $(\mathcal{F}_n)$ is the completion of a coherent $\mathcal{O}_U$-module.*

*Proof.* We will prove this by verifying hypotheses (a), (b), and (c) of Lemma 2.7.

Since $\mathcal{F}_n$ is locally free over $\mathcal{O}_U/f^n\mathcal{O}_U$ we see that we have short exact sequences $0 \to \mathcal{F}_n \to \mathcal{F}_{n+1} \to \mathcal{F}_1 \to 0$ for all $n$. Thus condition (b) holds by Cohomology, Lemma 0EHA.

As $f$ is a nonzerodivisor we obtain short exact sequences

$$0 \to A/f^nA \xrightarrow{f} A/f^{n+1}A \to A/fA \to 0$$

and we have corresponding short exact sequences $0 \to \mathcal{F}_n \to \mathcal{F}_{n+1} \to \mathcal{F}_1 \to 0$. We will use Local Cohomology, Lemma 0BK0 without further mention. Our assumptions imply that $H^0(U,\mathcal{O}_U/f\mathcal{O}_U)$ and $H^1(U,\mathcal{O}_U/f\mathcal{O}_U)$ are finite $A$-modules.

Hence the same thing is true for $\mathcal{F}_1$, see Local Cohomology, Lemma 0BLT. Using induction and the short exact sequences we find that $H^0(U, \mathcal{F}_n)$ are finite $A$-modules for all $n$. In this way we see hypothesis (c) is satisfied.

Finally, as $H^1(U, \mathcal{F}_1)$ is a finite $A$-module we can apply Cohomology, Lemma 0BLC to see hypothesis (a) holds. □

*Remark* 2.9. In Lemma 2.8 if $A$ is universally catenary with Cohen-Macaulay formal fibres (for example if $A$ has a dualizing complex), then the condition that $H^1_{\mathfrak{a}}(A/fA)$ and $H^2_{\mathfrak{a}}(A/fA)$ are finite $A$-modules, is equivalent with

$$\operatorname{depth}((A/f)_{\mathfrak{p}}) + \dim((A/\mathfrak{p})_{\mathfrak{q}}) > 2$$

for all $\mathfrak{p} \in V(f) \setminus V(\mathfrak{a})$ and $\mathfrak{q} \in V(\mathfrak{p}) \cap V(\mathfrak{a})$ by Local Cohomology, Theorem 0BJV.

For example, if $A/fA$ is $(S_2)$ and if every irreducible component of $Z = V(\mathfrak{a})$ has codimension $\geq 3$ in $Y = \operatorname{Spec}(A/fA)$, then we get the finiteness of $H^1_{\mathfrak{a}}(A/fA)$ and $H^2_{\mathfrak{a}}(A/fA)$. This should be contrasted with the slightly weaker conditions found in Lemma 0DXU (see also Remark 0DXV).

*Proof.* We will check hypotheses (a), (b), and (c) of Lemma 2.7. Before we start, let us point out that the modules $H^0(U, \mathcal{F}''_n)$ and $H^1(U, \mathcal{F}''_n)$ are finite $A$-modules for all $n$ by Local Cohomology, Lemma 0BJY.

Observe that for each $p \geq 0$ the limit topology on $\lim H^p(U, \mathcal{F}_n)$ is the $I$-adic topology by Lemma 2.5. In particular, hypothesis (b) holds.

We know that $M = \lim H^0(U, \mathcal{F}_n)$ is an $A$-module whose limit topology is the $I$-adic topology. Thus, given $n$, the module $M/I^nM$ is a subquotient of $H^0(U, \mathcal{F}_N)$ for some $N \gg n$. Since the inverse system $\{H^0(U, \mathcal{F}_N)\}$ is pro-isomorphic to an inverse system of finite $A$-modules, namely $\{H^0(U, \mathcal{F}''_N)\}$, we conclude that $M/I^nM$ is finite. It follows that $M$ is finite, see Algebra, Lemma 031D. In particular hypothesis (c) holds.

For each $n \geq 0$ let us write $Ob_n = \lim_N H^1(U, I^n\mathcal{F}_N)$. A special case is $Ob = Ob_0 = \lim_N H^1(U, \mathcal{F}_N)$. Arguing exactly as in the previous paragraph we find that $Ob$ is a finite $A$-module. (In fact, we also know that $Ob/IOb$ is annihilated by a power of $\mathfrak{a}$, but it seems somewhat difficult to use this.)

We set $\mathcal{F} = \lim \mathcal{F}_n$, we pick generators $f_1, \ldots, f_r$ of $I$, we pick $c \geq 1$, and we choose $\Phi_{\mathcal{F}}$ as in Lemma 2.4. We will use the results of Lemma 0EI8 without further mention. In particular, for each $n \geq 1$ there are maps

$$\delta_n : H^0(U, \mathcal{F}_n) \longrightarrow H^1(U, I^n\mathcal{F}) \longrightarrow Ob_n$$

The first comes from the short exact sequence $0 \to I^n\mathcal{F} \to \mathcal{F} \to \mathcal{F}_n \to 0$ and the second from $I^n\mathcal{F} = \lim I^n\mathcal{F}_N$. We will later use that if $\delta_n(s) = 0$ for $s \in H^0(U, \mathcal{F}_n)$ then we can for each $n' \geq n$ find $s' \in H^0(U, \mathcal{F}_{n'})$ mapping to $s$. Observe that there are commutative diagrams

$$\begin{array}{ccc}
H^0(U, \mathcal{F}_{nc}) & \longrightarrow & H^1(U, I^{nc}\mathcal{F}) \\
\downarrow & & \downarrow \\
H^0(U, \mathcal{F}_n) & \longrightarrow & H^1(U, I^n\mathcal{F})
\end{array}
\qquad
\begin{array}{ccc}
H^1(U, I^{nc}\mathcal{F}) & & \\
& \searrow^{\Phi_{\mathcal{F}}} & \\
& & \bigoplus_{e_1+\ldots+e_r=n} H^1(U, \mathcal{F}) \cdot T_1^{e_1} \ldots T_r^{e_r} \\
& \swarrow & \\
H^1(U, I^n\mathcal{F}) & &
\end{array}$$

We conclude that the obstruction map $H^0(U, \mathcal{F}_n) \to Ob_n$ sends the image of $H^0(U, \mathcal{F}_{nc}) \to H^0(U, \mathcal{F}_n)$ into the submodule

$$Ob'_n = \mathrm{Im}\left(\bigoplus\nolimits_{e_1+\ldots+e_r=n} Ob \cdot T_1^{e_1} \ldots T_r^{e_r} \to Ob_n\right)$$

where on the summand $Ob \cdot T_1^{e_1} \ldots T_r^{e_r}$ we use the map on cohomology coming from the reductions modulo powers of $I$ of the multiplication map $f_1^{e_1} \ldots f_r^{e_r} : \mathcal{F} \to I^n\mathcal{F}$. By construction

$$\bigoplus\nolimits_{n\geq 0} Ob'_n$$

is a finite graded module over the Rees algebra $\bigoplus_{n\geq 0} I^n$. For each $n$ we set

$$M_n = \{s \in H^0(U, \mathcal{F}_n) \mid \delta_n(s) \in Ob'_n\}$$

Observe that $\{M_n\}$ is an inverse system and that $f_j : \mathcal{F}_n \to \mathcal{F}_{n+1}$ on global sections maps $M_n$ into $M_{n+1}$. By exactly the same argument as in the proof of Cohomology, Lemma 0GYK we find that $\{M_n\}$ is ML. Namely, because the Rees algebra is Noetherian we can choose a finite number of homogeneous generators of the form $\delta_{n_j}(z_j)$ with $z_j \in M_{n_j}$ for the graded submodule $\bigoplus_{n\geq 0} \mathrm{Im}(M_n \to Ob'_n)$. Then if $k = \max(n_j)$ we find that for $n \geq k$ and any $z \in M_n$ we can find $a_j \in I^{n-n_j}$ such that $z - \sum a_j z_j$ is in the kernel of $\delta_n$ and hence in the image of $M_{n'}$ for all $n' \geq n$ (because the vanishing of $\delta_n$ means that we can lift $z - \sum a_j z_j$ to an element $z' \in H^0(U, \mathcal{F}_{n'c})$ for all $n' \geq n$ and then the image of $z'$ in $H^0(U, \mathcal{F}_{n'})$ is in $M_{n'}$ by what we proved above). Thus $\mathrm{Im}(M_n \to M_{n-k}) = \mathrm{Im}(M_{n'} \to M_{n-k})$ for all $n' \geq n$.

Choose $n$. By the Mittag-Leffler property of $\{M_n\}$ we just established we can find an $n' \geq n$ such that the image of $M_{n'} \to M_n$ is the same as the image of $M' \to M_n$. By the above we see that the image of $M' \to M_n$ contains the image of $H^0(U, \mathcal{F}_{n'c}) \to H^0(U, \mathcal{F}_n)$. Thus we see that $\{M_n\}$ and $\{H^0(U, \mathcal{F}_n)\}$ are pro-isomorphic. Therefore $\{H^0(U, \mathcal{F}_n)\}$ has ML and we finally conclude that hypothesis (a) holds. This concludes the proof. □

## References